

Project

OOP Java Tutorial-2

C:\Users\acer\OneDrive\Documents\Idea...

> .idea

> out

> Src

> SmartAttendanceChecker.java

External Libraries

Scratches and Consoles

SmartAttendanceChecker.java

```
1 import java.util.Scanner;
2
3 class AttendanceChecker {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter total classes conducted: ");
8         int totalClasses = scanner.nextInt();
9
10        System.out.print("Enter classes attended: ");
11        int attendedClasses = scanner.nextInt();
12
13        if (totalClasses <= 0) {
14            System.out.println("Total classes must be greater than 0.");
15        } else if (attendedClasses < 0 || attendedClasses > totalClasses) {
16            System.out.println("Invalid attendance value.");
17        } else {
18            double percentage = ((double) attendedClasses / totalClasses) * 100;
19            System.out.printf("Attendance: %.2f%%\n", percentage);
20        }
21    }
22 }
```

Run AttendanceChecker

Enter total classes conducted: 50

Enter classes attended: 25

Attendance: 50.00%

Status: Not Eligible

Process finished with exit code 0

Performance

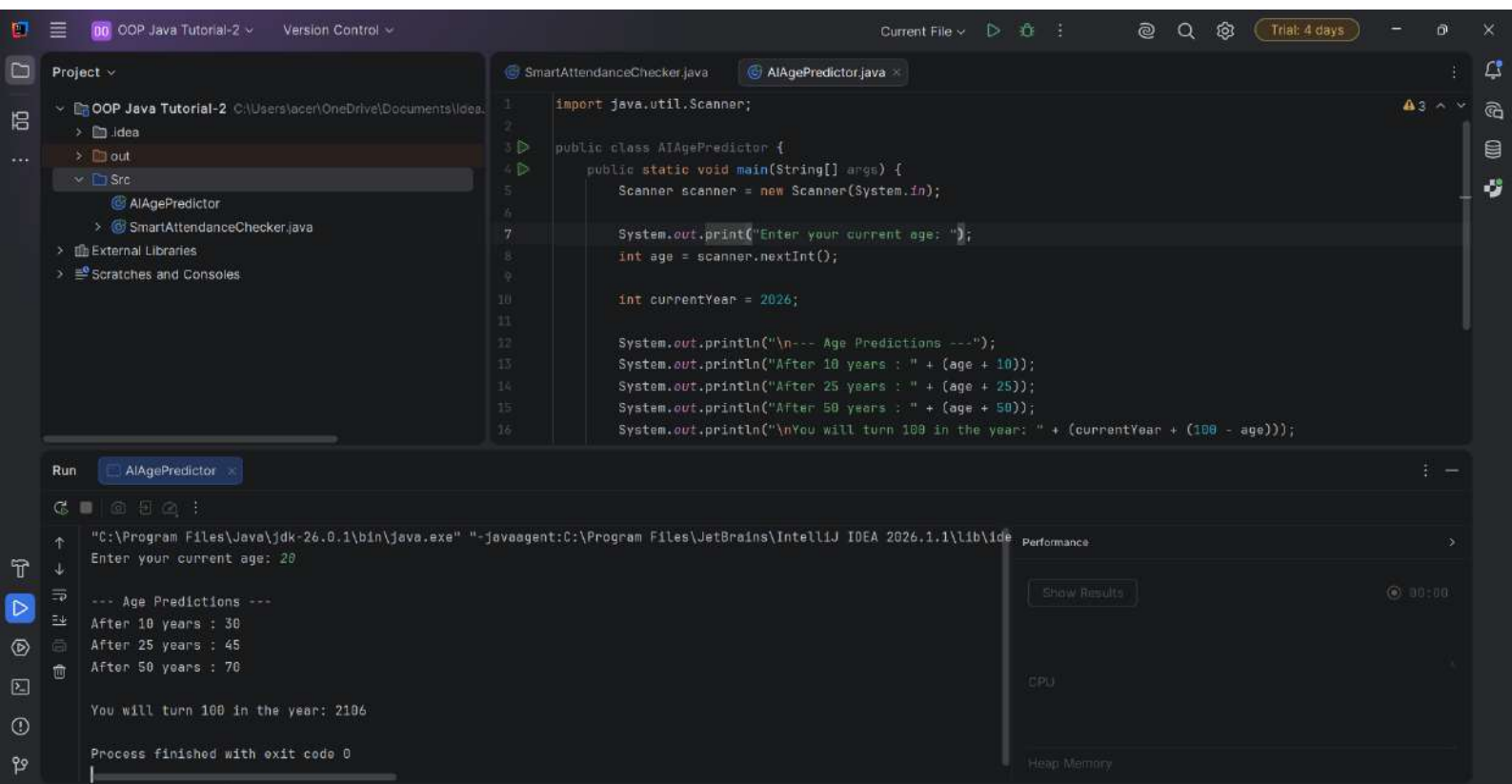
Show Results

CPU

Heap Memory

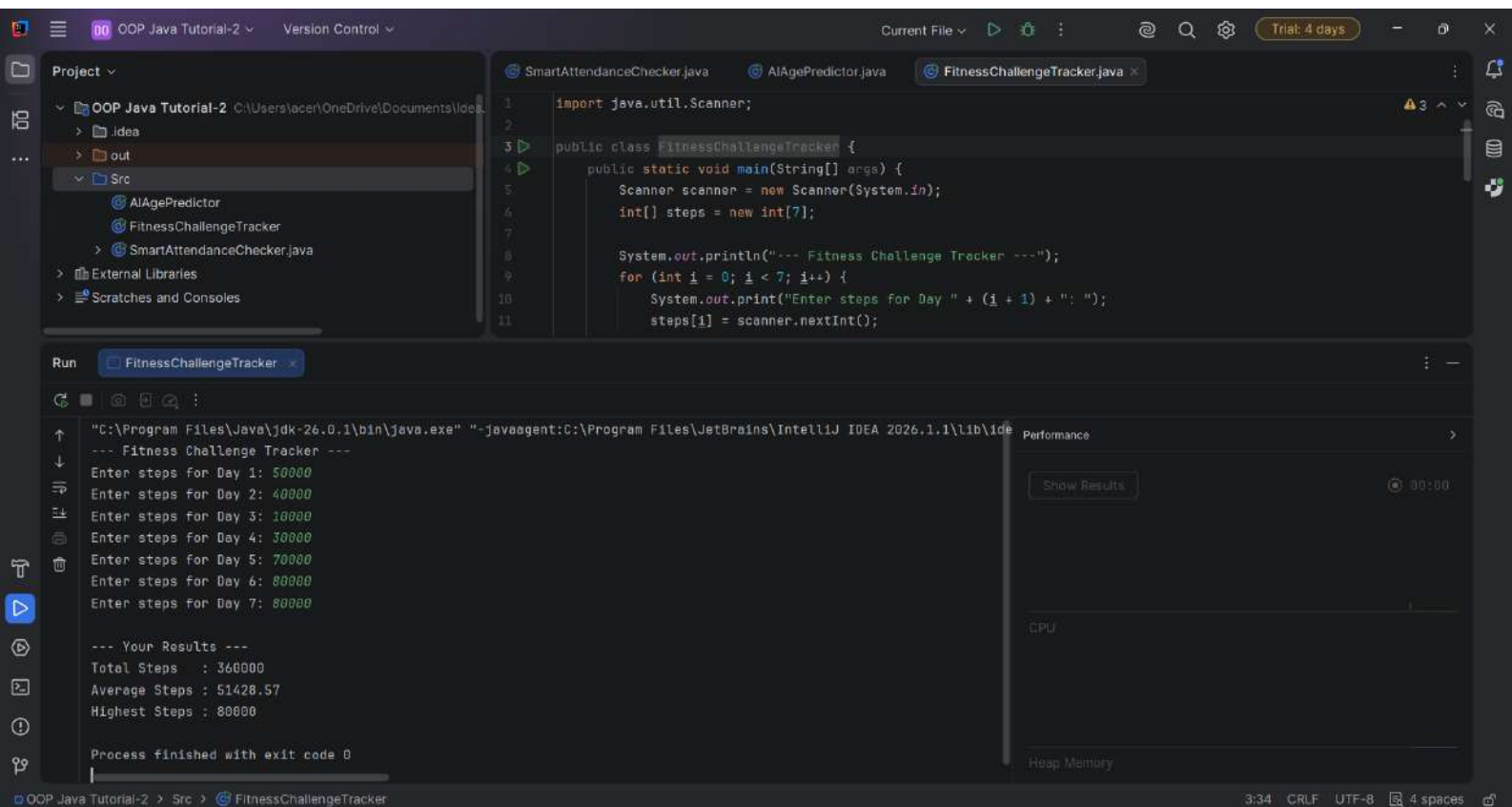
1) Smart Attendance checker

```
import java.util.Scanner;
class AttendanceChecker {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter total classes conducted:");
        int totalClasses = scanner.nextInt();
        System.out.print("Enter classes attended:");
        int attendedClasses = scanner.nextInt();
        if (totalClasses <= 0) {
            System.out.println("Total classes must be greater than 0.");
        } else if (attendedClasses < 0 || attendedClasses > totalClasses) {
            System.out.println("Invalid attendance value.");
        } else {
            double percentage = ((double) attendedClasses / totalClasses) * 100;
            System.out.printf("Attendance: %.2f%%\n", percentage);
            if (percentage >= 75) {
                System.out.println("Status: Eligible for exam ✓");
            } else {
                System.out.println("Status: Not Eligible ✗");
            }
        }
        scanner.close();
    }
}
```



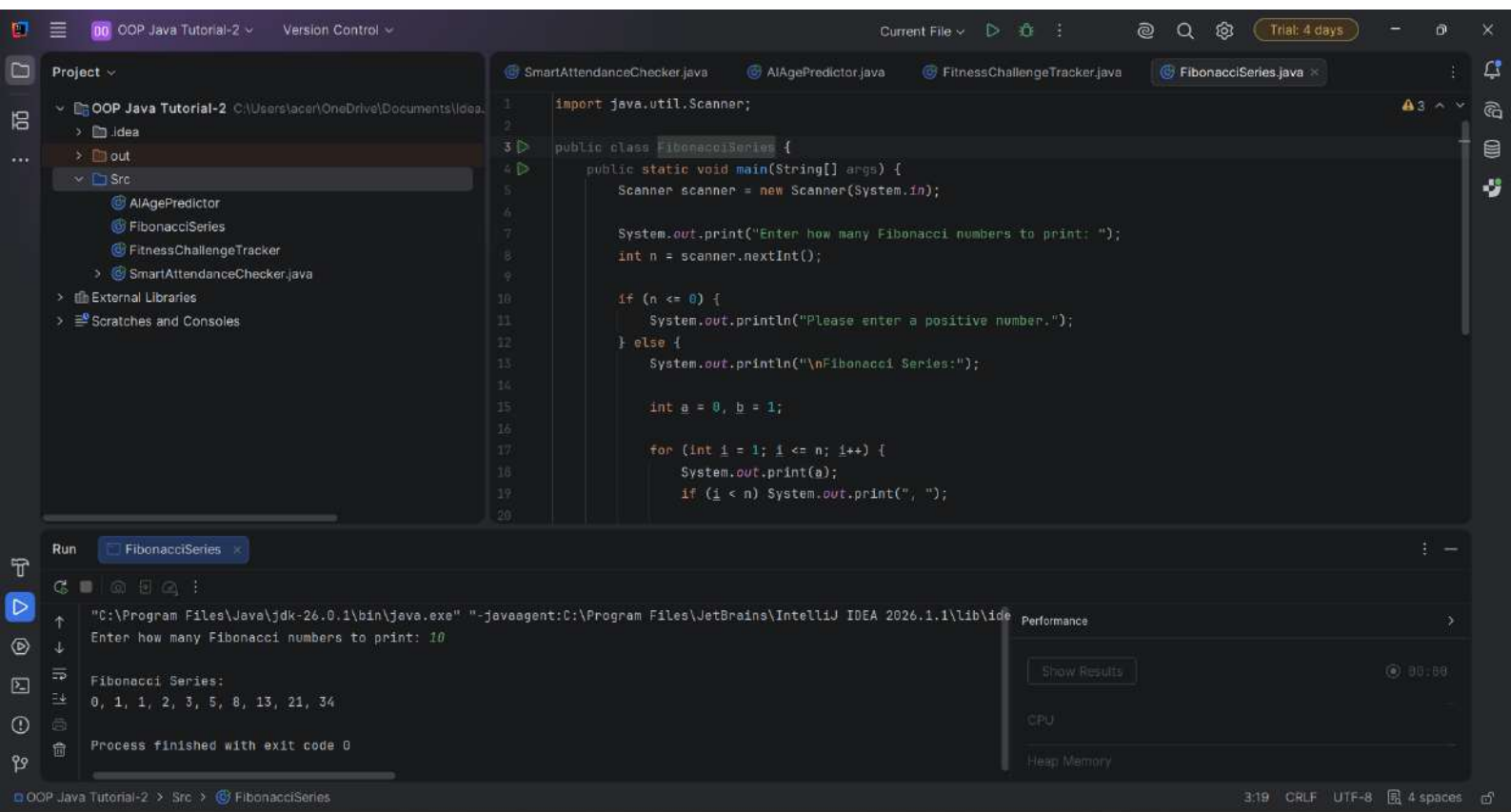
2) AI Age Predictor

```
import java.util.Scanner;
public class AIAgepredictor {
    public static void main (String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter your current age:");
        int age = scanner.nextInt();
        int currentYear = 2026;
        System.out.println("\n--- Age predictions ---");
        System.out.println("After 10 years: " + (age + 10));
        System.out.println("After 25 years: " + (age + 25));
        System.out.println("After 50 years: " + (age + 50));
        System.out.println("In you will turn 100 in the  
year: " + (currentYear + (100 - age)));
        scanner.close();
    }
}
```

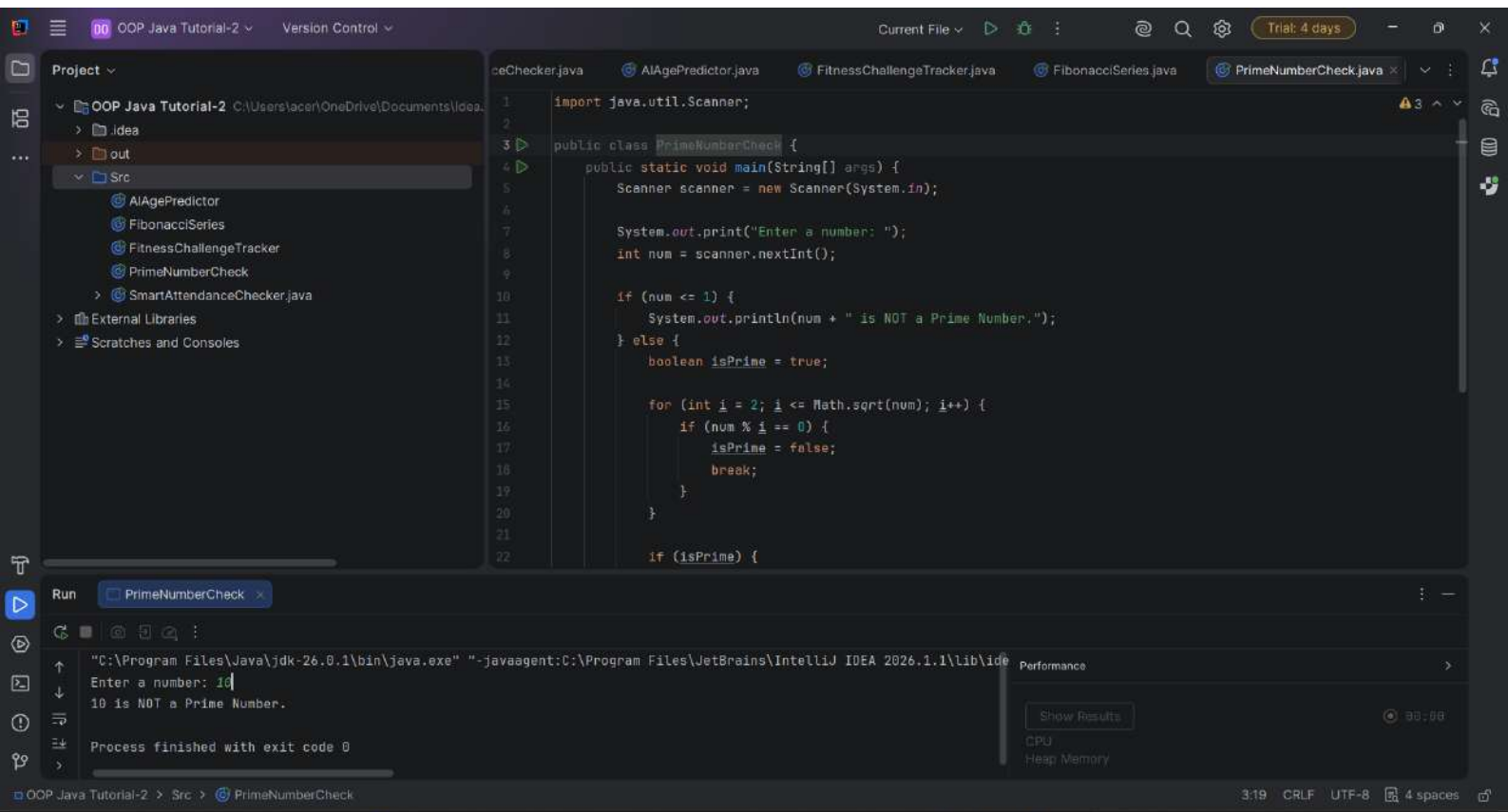
3) Fitness Challenge Tracker

```
import java.util.Scanner;
public class FitnessChallengeTracker {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int[] steps = new int[7];
        System.out.println("---fitness challenge tracker---");
        for (int i = 0; i < 7; i++) {
            System.out.print("Enter steps for day" + (i + 1)
                             + ": ");
            steps[i] = scanner.nextInt();
        }
        int total = 0;
        int highest = steps[0];
        for (int i = 0; i < 7; i++) {
            total += steps[i];
            if (steps[i] > highest) {
                highest = steps[i];
            }
        }
        double average = (double) total / 7;
        System.out.println("\n--- Your Results ---");
        System.out.println("Total steps : " + total);
        System.out.printf("Average step: %.2f\n", average);
        System.out.println("Highest step: " + highest);
        scanner.close();
    }
}
```



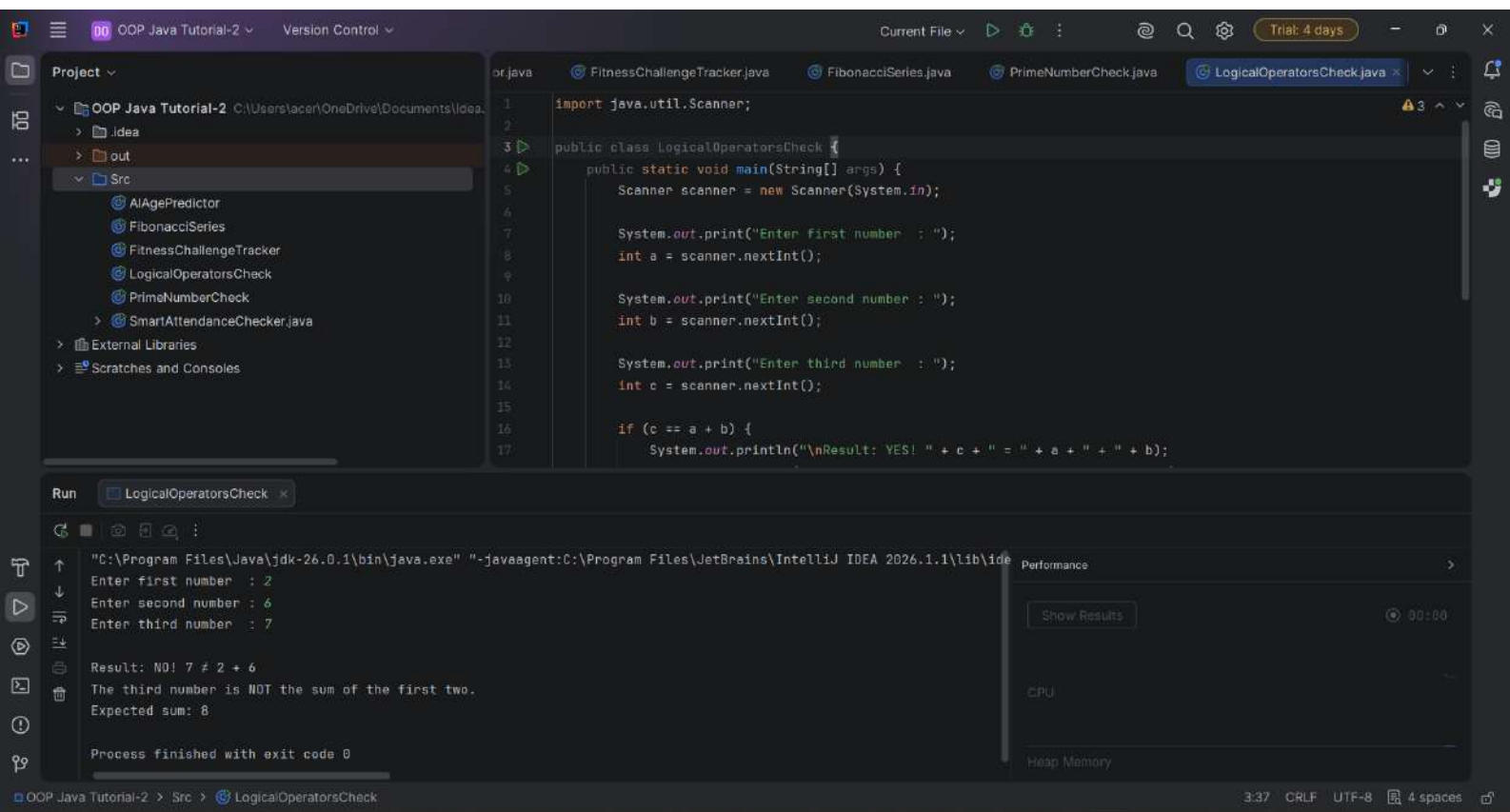
4) fibonacciseries

```
import java.util.Scanner;
public class fibonacciSeries {
    public static void main (String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.println("Enter how many fibonacci
            numbers to print:");
        int n = scanner.nextInt();
        if (n < 0) {
            System.out.println("please enter a positive
                number.");
        } else {
            System.out.println("In fibonacci series:");
            int a = 0, b = 1;
            for (int i = 1; i <= n; i++) {
                System.out.print(a);
                if (i < n) System.out.print(", ");
                int next = a + b;
                a = b;
                b = next;
            }
            System.out.println();
        }
        scanner.close();
    }
}
```

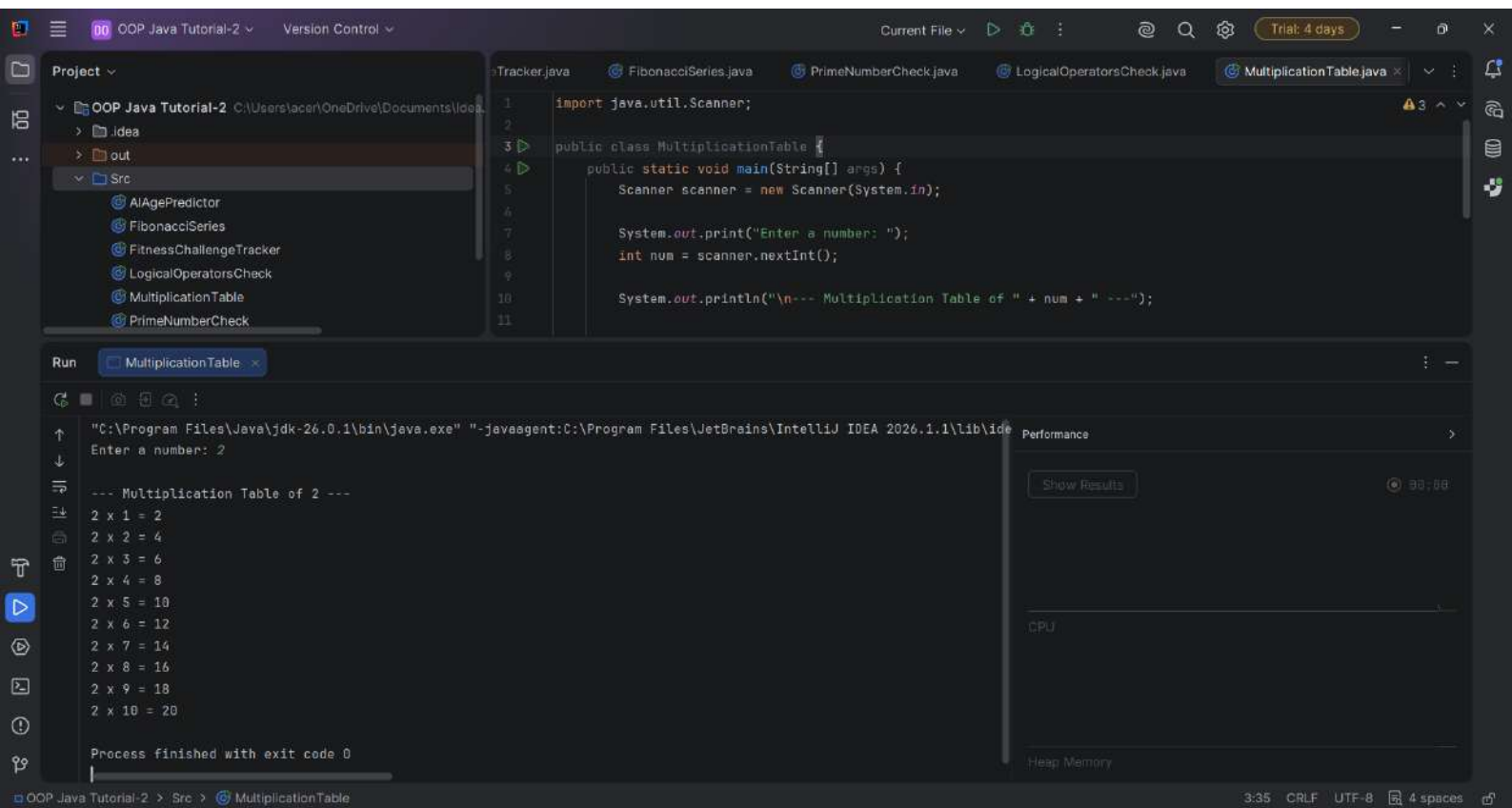
5) Prime number

```
import java.util.Scanner;
public class PrimeNumberCheck {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number:");
        int num = scanner.nextInt();
        if (num <= 1) {
            System.out.println(num + " is NOT a Prime Number");
        }
        else {
            boolean isPrime = true;
            for (int i = 2; i <= Math.sqrt(num); i++) {
                if (num % i == 0) {
                    isPrime = false;
                    break;
                }
            }
            if (isPrime) {
                System.out.println(num + " is a Prime Number");
            }
            else {
                System.out.println(num + " is NOT a Prime number");
            }
        }
        scanner.close();
    }
}
```



6) Logical operators check

```
import java.util.Scanner;
public class LogicalOperatorCheck {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter first number:");
        int a = scanner.nextInt();
        System.out.print("Enter second number:");
        int b = scanner.nextInt();
        System.out.print("Enter third number:");
        int c = scanner.nextInt();
        if (c == a + b) {
            System.out.println("In Result: Yes! " + c + " = " +
                               a + " + " + b);
            System.out.println("The third number is the  
- sum of the first two.");
        } else {
            System.out.println("In Result: No! " + c + " ≠ " + a +
                               " + " + b);
            System.out.println("The third number is  
NOT the sum of the first two.");
            System.out.println("Expected sum: " + (a + b));
        }
        scanner.close();
    }
}
```

7) Multiplication Table

```
import java.util.Scanner;
public class multiplicationTable {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number: ");
        System.out.println("\n --- Multiplication Table  
of " + num + " ---");
        for (int i = 1; i <= 10; i++) {
            System.out.println(num + "x" + i + " = " + (num * i));
        }
        scanner.close();
    }
}
```


8) Right Angled Triangle Pattern

```
import java.util.Scanner;
```

```
public class RightAngledTrianglePattern {
```

```
    public static void main(String[] args) {
```

```
        Scanner scanner = new Scanner(System.in);
```

```
        System.out.print("Enter number of rows:");
```

```
        int rows = scanner.nextInt();
```

```
        System.out.println("In -- Right Angled Triangle  
        ---");
```

```
        for (int i = 1; i <= rows; i++) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print("*");
```

```
            }
```

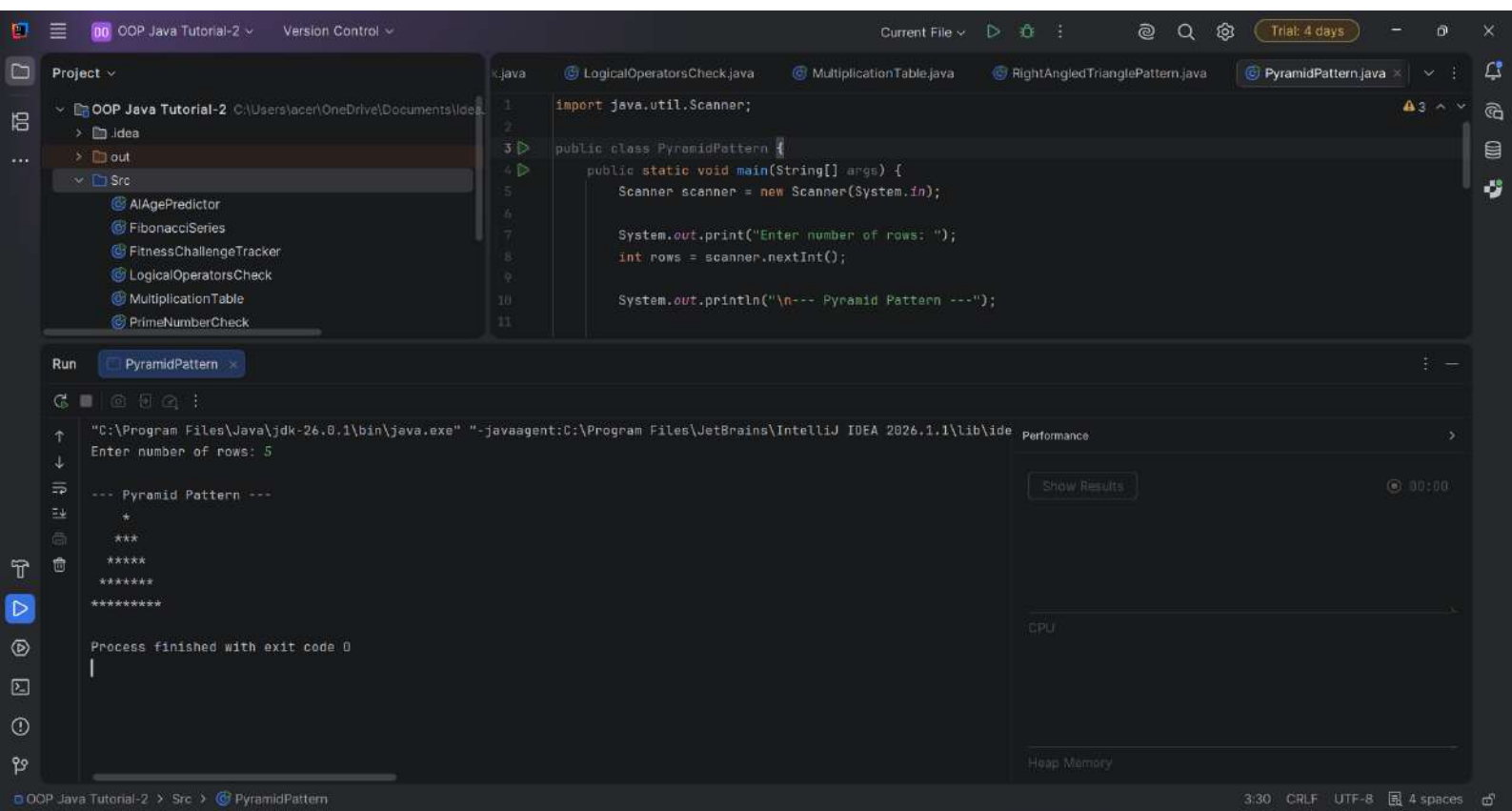
```
            System.out.println();
```

```
        }
```

```
        scanner.close();
```

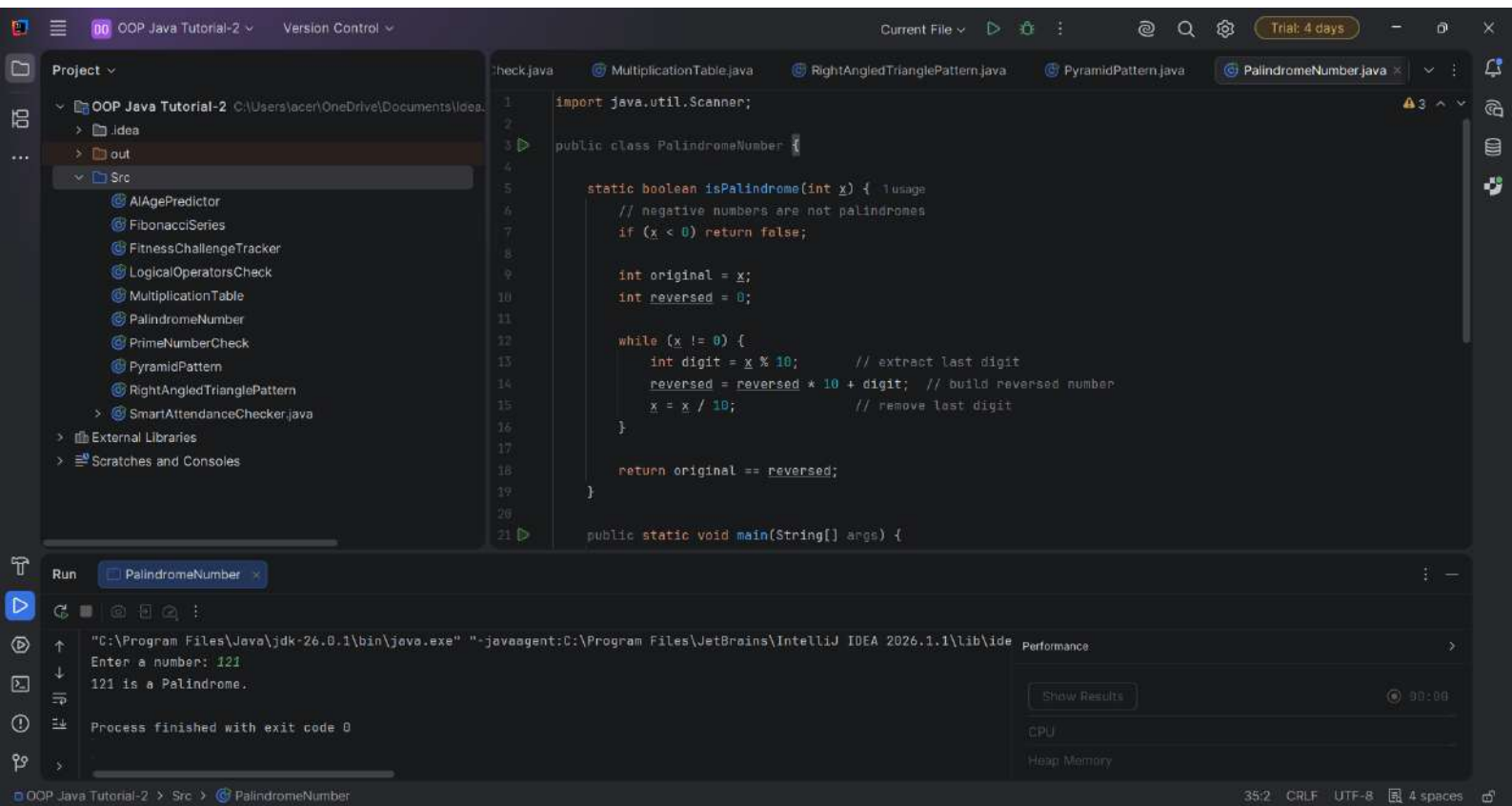
```
    }
```

```
}
```

9) Pyramid Pattern

```
import java.util.Scanner;
public class PyramidPattern {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter number of rows:");
        int rows = scanner.nextInt();
        System.out.println("In ---Pyramid Pattern---");
        for (int i = 1; i <= rows; i++) {
            // Print spaces
            for (int j = 1; j < rows; j++) {
                System.out.print(" ");
            }
            // Print stars
            for (int k = 1; k <= (2 * i - 1); k++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

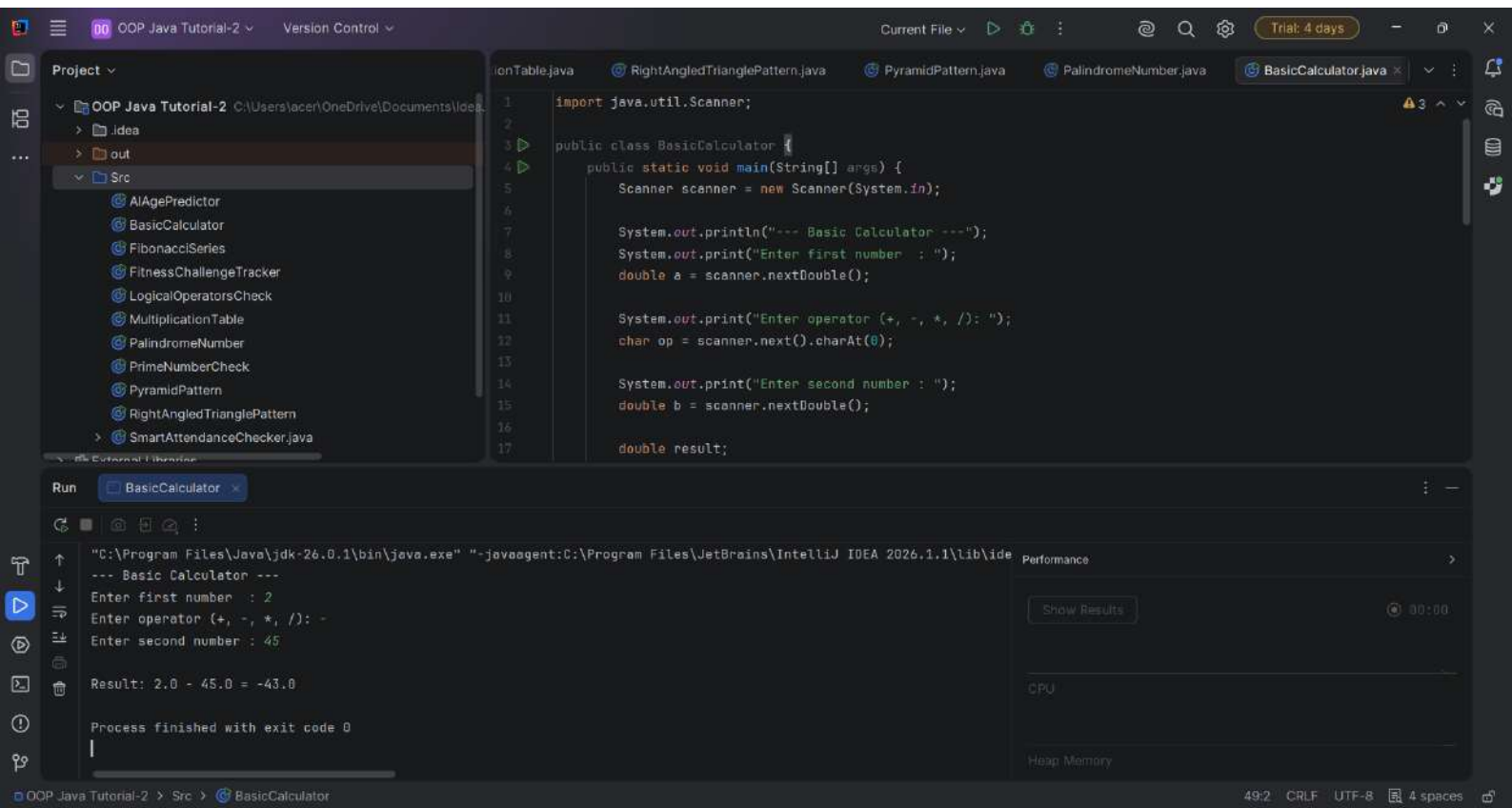


10) Palindrome number

```
import java.util.Scanner;
public class PalindromeNumber {
    static boolean isPalindrome(int x) { // usage
        if (x < 0) return false;
        int original = x;
        int reversed = 0;
        while (x != 0) {
            int digit = x % 10;
            reversed = reversed * 10 + digit;
            x = x / 10;
        }
    }
}
```

```
return original == reversed;
}
```

```
public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);
    System.out.print("Enter a number: ");
    int num = scanner.nextInt();
    if (isPalindrome(num)) {
        System.out.println(num + " is a palindrome.");
    } else {
        System.out.println(num + " is NOT a palindrome.");
    }
    scanner.close();
}
```

11) Basic calculator

```
import java.util.Scanner
public class Basiccalculator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in)
        System.out.print("--- Basic calculator ---");
        System.out.print("Enter first number:");
        double a = scanner.nextDouble();
        System.out.print("Enter operator (+, -, *, /):");
        char op = scanner.next().charAt(0);
        System.out.print("Enter second number:");
        double b = scanner.nextDouble();
        double result;
        String operation;
        switch (op) {
            case '+':
                result = a + b;
                operation = a + "+" + b + "=" + result;
                break;
            case '-':
                result = a - b;
                operation = a + "-" + b + "=" + result;
                break;
            case '*':
                result = a * b;
                operation = a + "*" + b + "=" + result;
                break;
            case '/':
                if (b == 0) {
```

system.out.println("Error, Division by zero!");
return;

}

result = a / b;

operation = a + "/" + b + "=" + result;

break;

default:

system.out.println("Error, Invalid
operator!");

return

}

system.out.println("In Result: " + operation);

scanner.close();

}

}

2) Grade calculator

```
import java.util.Scanner;
public class GradeCalculator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter marks obtained (0-100): ");

        double marks = scanner.nextDouble();
        if (marks < 0 || marks > 100) {
            System.out.println("Invalid marks! Please enter between 0 and 100.");
        } else {
            String grade = marks >= 90 ? "A+" :
                marks >= 85 ? "A" :
                marks >= 80 ? "A-" :
                marks >= 75 ? "B+" :
                marks >= 70 ? "B" :
                marks >= 65 ? "B-" :
                marks >= 60 ? "C+" :
                marks >= 55 ? "C" :
                marks >= 50 ? "C-" :
                marks >= 45 ? "D+" :
                marks >= 40 ? "D" : "F";

            String remarks = marks >= 80 ? "Distinction" :
                marks >= 65 ? "Credit" :
                marks >= 50 ? "Pass" :
                marks >= 40 ? "Conceded Pass" : "Fail";

            double gpa = marks >= 90 ? 4.0 :
                marks >= 85 ? 3.7 :
```

marks >= 80 ? 3.3:

marks >= 75 ? 3.0:

marks >= 70 ? 2.7:

marks >= 65 ? 2.3:

marks >= 55 ? 1.7:

marks >= 50 ? 1.3:

marks >= 45 ? 1.0:

marks >= 40 ? 0.7 : 0.0:

system.out.println("In ---Result ---");

system.out.println("marks: "+marks);

system.out.println("marks: "+grade);

system.out.println("marks: "+gpa);

system.out.println(~~"marks: "~~ ("Remarks:
"+remarks);

}

scanner.close();

}

}